

Mapping and Monitoring Surface Water Features in Alaska Through Machine Learning

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Land use practices and ecological conditions are largely influenced by the availability of freshwater in an area. In cold regions such as Alaska, hydrologic conditions are particularly important due to the varying terrain types and their susceptibility to freeze-thaw cycles. To update the National Map data, the United States Geological Survey (USGS) is exploring automated machine learning (ML) techniques to extract surface water features from high-resolution (HR) remotely-sensed elevation and image data. The goal is to support the validation or updating of the vector hydrography database for the 3D Hydrography Program (3DHP) and to monitor changes in these features as new remote sensing data become available. Here we report progress for Alaska.

Methods employ U-net neural network models with transfer learning to extract hydrographic features based on best available vector training data (i.e. reference hydrography). Using traditional flow-routing techniques and guidance from geomorphic indicators, reference hydrography is derived from 5-meter resolution elevation models produced from Interferometric Synthetic Aperture Radar data. Subsequent manual editing aligns derived reference features with HR image data. Using reference hydrography for approximately 25 percent of a 4600 square kilometer (sq km) study area to train a U-net base model, the model achieves an average F1 score of 80 percent for test areas. The base model is then fine-tuned using training data for roughly 6 percent of additional target regions in Alaska to predict hydrography for these regions.

Vector features are generated from raster U-net predictions. Network line features are derived from d-infinity flow lines that are constrained to follow the U-net predictions, and vector waterbodies are derived from predictions through contraction-expansion buffer operations. Preliminary results for target areas show average F1 scores of roughly 70 percent for vectorized predictions when compared to reference hydrography. Model prediction uncertainties are estimated using Monte Carlo simulation and precision metrics derived for the input data layers. Results indicate that ML has the potential to automate update and validation of elevation-derived hydrography from remote sensing data. Ongoing research is extending this work by testing ML methods to automate HR image alignment and hydrography feature extraction, with the goal of monitoring temporal changes in surface water extent.

Keywords: hydrography, deep learning, transfer learning, feature extraction, rivers

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